HDOC Day-66

Question: Write a recursive program to multiply two square matrices. Do not use iterative loops for the multiplication logic.

C Program

```
#include <stdio.h>
#define MAX 10

void multiply(int n, int A[MAX][MAX], int B[MAX][MAX],
              int C[MAX][MAX], int i, int j, int k) {
    if (i == n) return;
    if (j == n) {
        multiply(n, A, B, C, i + 1, 0, 0);
        return;
    }
    if (k == n) {
        multiply(n, A, B, C, i, j + 1, 0);
        return;
    }
    C[i][j] += A[i][k] * B[k][j];
    multiply(n, A, B, C, i, j, k + 1);
}

int main() {
    int n, A[MAX][MAX], B[MAX][MAX], C[MAX][MAX] = {0};
    printf("Enter matrix size (1-10): ");
    scanf("%d", &n);
    if (n < 1 || n > MAX) return 1;

    printf("Enter elements of matrix A:\n");
    for (int i = 0; i < n; i++)
        for (int j = 0; j < n; j++)
            scanf("%d", &A[i][j]);

    printf("Enter elements of matrix B:\n");
    for (int i = 0; i < n; i++)
        for (int j = 0; j < n; j++)
            scanf("%d", &B[i][j]);

    multiply(n, A, B, C, 0, 0, 0);
    printf("Product matrix:\n");
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++)
            printf("%d ", C[i][j]);
        printf("\n");
    }
    return 0;
}
```

Sample Input and Output

```
Enter matrix size (1-10): 2
Enter elements of matrix A:
1 2
3 4
Enter elements of matrix B:
5 6
7 8
Product matrix:
19 22
```

```
43 50
```

Explanation

The recursive function calculates each element of the product matrix by adding A[i][k] * B[k][j]. The index k moves across the shared dimension, j moves to the next column, and i moves to the next row. No loops are used in the multiplication logic; loops are used only for input and output.